

ConvexOptimization 2

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PROBLEM I Definition of convexity.

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex, and $a, b \in \text{dom } f$ with $a < b$.

1. Show that

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b)$$

for all $x \in [a, b]$.

2. Show that

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}$$

for all $x \in (a, b)$. Draw a sketch that illustrates this inequality.

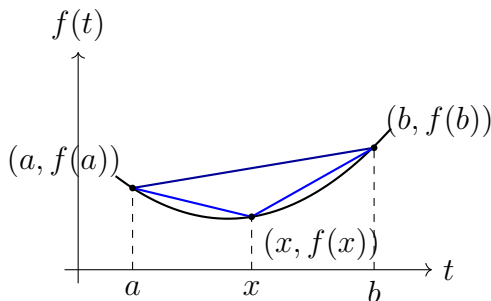
3. Suppose f is differentiable. Use the result in (b) to show that

$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b).$$

4. Suppose f is twice differentiable. Use the result in (c) to show that $f''(a) \geq 0$ and $f''(b) \geq 0$.

SOLUTION. 1. 令 $\theta = (x - a)/(b - a)$ 。当 $x \in [a, b]$ 时, $\theta \in [0, 1]$, 且 $x = (1 - \theta)a + \theta b$ 。由凸性, $f(x) \leq (1 - \theta)f(a) + \theta f(b) = \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b)$ 。

2. 由上一问, $f(x) - f(a) \leq \frac{x-a}{b-a}(f(b) - f(a))$, 同理, $f(b) - f(x) \geq \frac{b-x}{b-a}(f(b) - f(a))$, 显然题目不等式成立。几何上, 凸函数图像上割线斜率随区间向右移动而不减, 因此 $[a, x]$ 、 $[a, b]$ 、 $[x, b]$ 三条割线的斜率依次不减。可用下图表示: 中间的长割线斜率介于左、右两条割线斜率之间。



3. 由于 $\frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(x)}{b-x}$, 那么 $\lim_{x \downarrow a} \frac{f(x)-f(a)}{x-a} = f'(a) \leq \lim_{x \downarrow a} \frac{f(b)-f(a)}{b-a}$. 同理有 $f'(b) \geq \frac{f(b)-f(a)}{b-a}$.
4. 对区间 $[a, a+h]$ 应用上一问, 有 $f'(a) \leq \frac{f(a+h)-f(a)}{h} \leq f'(a+h)$. 故 $0 \leq (f'(a+h) - f'(a))/h$, 令 $h \downarrow 0$ 得 $f''(a) \geq 0$. 同理, 对 $[b-h, b]$ 应用上一问并令 $h \downarrow 0$, 得 $f''(b) \geq 0$.

□

PROBLEM II An extension of Jensen's inequality.

One interpretation of Jensen's inequality is that randomization or dithering hurts, i.e., raises the average value of a convex function: For f convex and v a zero mean random variable, we have $\mathbf{E}f(x_0 + v) \geq f(x_0)$. This leads to the following conjecture. If f_0 is convex, then the larger the variance of v , the larger $\mathbf{E}f(x_0 + v)$.

1. Give a counterexample that shows that this conjecture is false. Find zero mean random variables v and w , with $\text{var}(v) > \text{var}(w)$, a convex function f , and a point x_0 , such that $\mathbf{E}f(x_0 + v) < \mathbf{E}f(x_0 + w)$.
2. The conjecture is true when v and w are scaled versions of each other. Show that $\mathbf{E}f(x_0 + tv)$ is monotone increasing in $t \geq 0$, when f is convex and v is zero mean.

SOLUTION. 1. 取 $x_0 = 0$, $f(x) = \max\{x, 0\}$. 令 w, v 两个随机变量, $\mathbb{P}(w = 1) = \mathbb{P}(w = -1) = \frac{1}{2}$, $\mathbb{P}(v = 20) = \frac{1}{100}$, $\mathbb{P}(v = -\frac{20}{99}) = \frac{99}{100}$. 则 $\mathbf{E}w = \mathbf{E}v = 0$, $\text{var}(w) = 1$, 且 $\text{var}(v) = \mathbf{E}v^2 = \frac{20^2}{100} + \frac{99}{100} \frac{20^2}{99^2} = \frac{400}{99} > 1$. 但 $\mathbf{E}f(v) = \frac{20}{100} = \frac{1}{5} < \frac{1}{2} = \mathbf{E}f(w)$. 因此该猜想不成立.

2. 记 $F(t) = \mathbf{E}f(x_0 + tv)$. 若 $0 \leq s \leq t$ 且 $t > 0$, 令 $\lambda = s/t$, 则 $x_0 + sv = (1-\lambda)x_0 + \lambda(x_0 + tv)$. 凸性与 Jensen 不等式给出 $F(s) \leq (1-\lambda)f(x_0) + \lambda F(t) \leq F(t)$, 因为 $F(t) \geq f(x_0 + \mathbf{E}tv) = f(x_0)$. 故 F 在 $t \geq 0$ 上单调不减.

□

PROBLEM III Suppose $p < 1$, $p \neq 0$. Show that the function

$$f(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}$$

with $\text{dom } f = \mathbb{R}_{++}^n$ is concave. This includes as special cases $f(x) = (\sum_{i=1}^n x_i^{1/2})^2$ and the harmonic mean $f(x) = (\sum_{i=1}^n 1/x_i)^{-1}$.

Hint. Adapt the proofs for the log-sum-exp function and the geometric mean in §3.1.5.

SOLUTION. 记 $S = \sum_i x_i^p$, 则 $f(x) = S^{1/p}$. 直接求二阶方向导数可得, 对任意 $z \in \mathbb{R}^n$,

$$z^T \nabla^2 f(x) z = (1-p) S^{1/p-2} \left[\left(\sum_i z_i x_i^{p-1} \right)^2 - S \sum_i z_i^2 x_i^{p-2} \right].$$

由 Cauchy-Schwarz 不等式,

$$\left(\sum_i z_i x_i^{p-1} \right)^2 = \left(\sum_i x_i^{p/2} z_i x_i^{p/2-1} \right)^2 \leq \left(\sum_i x_i^p \right) \left(\sum_i z_i^2 x_i^{p-2} \right).$$

因此方括号内非正。又 $p < 1$, 故 $1 - p > 0$, 从而 $z^T \nabla^2 f(x) z \leq 0$ 。所以 $\nabla^2 f(x) \preceq 0$, f 为凹函数。□

PROBLEM IV Nonnegative weighted sums and integrals.

1. Show that $f(x) = \sum_{i=1}^r \alpha_i x_{[i]}$ is a convex function of x , where $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_r \geq 0$, and $x_{[i]}$ denotes the i th largest component of x . (You can use the fact that $f(x) = \sum_{i=1}^k x_{[i]}$ is convex on $\mathbb{R}^n$.)
2. Let $T(x, \omega)$ denote the trigonometric polynomial

$$T(x, \omega) = x_1 + x_2 \cos \omega + x_3 \cos 2\omega + \cdots + x_n \cos(n-1)\omega.$$

Show that the function

$$f(x) = - \int_0^{2\pi} \log T(x, \omega) d\omega$$

is convex on $\{x \in \mathbb{R}^n \mid T(x, \omega) > 0, 0 \leq \omega \leq 2\pi\}$.

SOLUTION. 1. 令 $S_k(x) = \sum_{i=1}^k x_{[i]}$ 。则

$$S_k(x) = \max \left\{ \sum_{i \in I} x_i \mid I \subseteq \{1, \dots, n\}, |I| = k \right\}.$$

事实上, 固定 x 后, 右端显然通过取 x 中最大的 k 个分量达到, 因而等于 $S_k(x)$ 。由于每个 $\sum_{i \in I} x_i$ 都是 x 的线性函数, 而有限个线性函数的点态最大值是凸函数, 所以 S_k 凸。

现在由 Abel 变换,

$$\sum_{i=1}^r \alpha_i x_{[i]} = \sum_{k=1}^{r-1} (\alpha_k - \alpha_{k+1}) S_k(x) + \alpha_r S_r(x).$$

由于 $\alpha_k - \alpha_{k+1} \geq 0$, $\alpha_r \geq 0$, 且每个 S_k 凸, 所以该函数凸。

2. 对固定 ω , $T(x, \omega)$ 是 x 的仿射函数, 且 $-\log u$ 在 $u > 0$ 上凸, 故 $x \mapsto -\log T(x, \omega)$ 凸。定义域也是凸的, 因为 $T(\theta x + (1-\theta)y, \omega) = \theta T(x, \omega) + (1-\theta)T(y, \omega) > 0$ 。凸函数的非负积分仍凸, 因此

$$f(x) = - \int_0^{2\pi} \log T(x, \omega) d\omega$$

是凸函数。□

PROBLEM V Convex hull or envelope of a function.

The convex hull or convex envelope of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as

$$g(x) = \inf\{t \mid (x, t) \in \text{conv epi } f\}.$$

Geometrically, the epigraph of g is the convex hull of the epigraph of f . Show that g is the largest convex underestimator of f . In other words, show that if h is convex and satisfies $h(x) \leq f(x)$ for all x , then $h(x) \leq g(x)$ for all x .

SOLUTION. 因为 $(x, f(x)) \in \text{epi } f \subseteq \text{conv epi } f$, 所以 $g(x) \leq f(x)$ 。又 $\text{conv epi } f$ 是凸且向上闭合的集合, 故 $\text{epi } g = \text{conv epi } f$, 从而 g 凸。

设 h 凸且 $h \leq f$ 。则 $\text{epi } f \subseteq \text{epi } h$ 。由于 $\text{epi } h$ 是凸集, $\text{conv epi } f \subseteq \text{epi } h$ 。因此若 $(x, t) \in \text{conv epi } f$, 则 $t \geq h(x)$ 。对所有这样的 t 取下确界, 得 $g(x) \geq h(x)$ 。故 g 是最大的凸下估计。□

PROBLEM VI Products and ratios of convex functions.

In general the product or ratio of two convex functions is not convex. However, there are some results that apply to functions on $\mathbb{R}$. Prove the following.

1. If f and g are convex, both nondecreasing (or nonincreasing), and positive functions on an interval, then fg is convex.
2. If f, g are concave, positive, with one nondecreasing and the other nonincreasing, then fg is concave.
3. If f is convex, nondecreasing, and positive, and g is concave, nonincreasing, and positive, then f/g is convex.

SOLUTION. 1. 设 $x \leq y$, $z = \theta x + (1 - \theta)y$, 并记 $A = f(x), B = f(y), C = g(x), D = g(y)$ 。若 f, g 同为非减或同为非增, 则 $(B - A)(D - C) \geq 0$ 。由凸性和正性, $f(z)g(z) \leq [\theta A + (1 - \theta)B][\theta C + (1 - \theta)D]$ 。而 $\theta AC + (1 - \theta)BD - [\theta A + (1 - \theta)B][\theta C + (1 - \theta)D] = \theta(1 - \theta)(B - A)(D - C) \geq 0$ 。故 fg 凸。

2. 不妨设 f 非减、 g 非增。沿用上面的记号, 则 $(B - A)(D - C) \leq 0$ 。由凹性和正性, $f(z)g(z) \geq [\theta A + (1 - \theta)B][\theta C + (1 - \theta)D]$ 。又 $[\theta A + (1 - \theta)B][\theta C + (1 - \theta)D] - \theta AC - (1 - \theta)BD = -\theta(1 - \theta)(B - A)(D - C) \geq 0$ 。故 fg 凹。若 f 非增、 g 非减, 则 $B - A \leq 0, D - C \geq 0$, 仍有 $(B - A)(D - C) \leq 0$, 上述同一计算完全适用, 因此这种单调性情形下 fg 也凹。

3. 因 g 正且凹, $q = 1/g$ 是凸函数; 又因 g 非增, q 非减。于是 f 与 q 都是正的、凸的、非减函数, 由第 (a) 问可知 $f/g = fq$ 凸。

□

PROBLEM VII Properties of conjugate functions.

1. *Conjugate of convex plus affine function.* Define $g(x) = f(x) + c^T x + d$, where f is convex. Express g^* in terms of f^* (and c, d).
2. *Conjugate of perspective.* Express the conjugate of the perspective of a convex function f in terms of f^* .
3. *Conjugate and minimization.* Let $f(x, z)$ be convex in (x, z) and define $g(x) = \inf_z f(x, z)$. Express the conjugate g^* in terms of f^* .

As an application, express the conjugate of $g(x) = \inf_z \{h(z) \mid Az + b = x\}$, where h is convex, in terms of h^* , A , and b .

4. *Conjugate of conjugate.* Show that the conjugate of the conjugate of a closed convex function is itself: $f = f^{**}$ if f is closed and convex. (A function is closed if its epigraph is closed; see §A.3.3.)

Hint. Show that f^{**} is the pointwise supremum of all affine global underestimators of f . Then apply the result of exercise 3.28.

SOLUTION. 1. $g^*(y) = \sup_x \{y^T x - f(x) - c^T x - d\} = f^*(y - c) - d$.

2. 设 $g(x, t) = t f(x/t)$, $t > 0$ 。令 $u = x/t$, 则 $g^*(y, s) = \sup_{t>0, u} t(y^T u + s - f(u))$ 。因而

$$g^*(y, s) = \begin{cases} 0, & f^*(y) + s \leq 0, \\ +\infty, & f^*(y) + s > 0. \end{cases}$$

3. $g^*(y) = \sup_x \sup_z \{y^T x - f(x, z)\} = f^*(y, 0)$ 。那么, $g^*(y) = \sup_z \{y^T (Az + b) - h(z)\} = b^T y + h^*(A^T y)$ 。

4. 对任意 a , 仿射函数 $a^T x + b$ 是 f 的全局下估计当且仅当 $b \leq -f^*(a)$ 。因此所有仿射全局下估计的点态上确界为 $\sup_a \{a^T x - f^*(a)\} = f^{**}(x)$ 。一方面, 由 $f^*(a) \geq a^T x - f(x)$ 得 $f^{**}(x) \leq f(x)$ 。另一方面, 若 f 闭且凸, 则由闭凸集 $\text{epi } f$ 的支撑超平面定理, 任意 $t < f(x)$ 都存在仿射全局下估计在 x 处严格大于 t 。令 $t \uparrow f(x)$, 得 $f^{**}(x) \geq f(x)$ 。故 $f^{**} = f$ 。

□

PROBLEM VIII First-order condition for quasiconvexity.

Prove the first-order condition for quasiconvexity given in §3.4.3: A differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, with $\text{dom } f$ convex, is quasiconvex if and only if for all $x, y \in \text{dom } f$,

$$f(y) \leq f(x) \implies \nabla f(x)^T (y - x) \leq 0.$$

Hint. It suffices to prove the result for a function on $\mathbb{R}$; the general result follows by restriction to an arbitrary line.

SOLUTION. 先证一维情形。若 ϕ 拟凸且 $\phi(y) \leq \phi(x)$, 则对 $t \in [0, 1]$,

$$\phi(x + t(y - x)) \leq \max\{\phi(x), \phi(y)\} = \phi(x).$$

令 $t \downarrow 0$ 得 $\phi'(x)(y - x) \leq 0$ 。

反过来, 若该条件成立但 ϕ 非拟凸, 则存在 $a < b < c$ 使 $\phi(b) > \max\{\phi(a), \phi(c)\}$ 。取 $\max\{\phi(a), \phi(c)\} < \alpha < \phi(b)$ 。由连续性和均值定理, 可在某点 s 找到 $\phi(s) > \alpha$ 且导数指向一个满足函数值不超过 α 的端点, 从而有 $\phi'(s)(y - s) > 0$, 与假设矛盾。因此 ϕ 拟凸。

多维情形限制到任意线段

$$\phi(t) = f(x + t(y - x))$$

即得结论, 因为 $\phi'(0) = \nabla f(x)^T (y - x)$, 而函数在凸定义域上拟凸等价于在每条线段上的限制均拟凸。

□

PROBLEM IX Second-order conditions for quasiconvexity.

In this problem we derive alternate representations of the second-order conditions for quasiconvexity given in §3.4.3. Prove the following.

1. A point $x \in \text{dom } f$ satisfies (3.21) if and only if there exists a σ such that

$$\nabla^2 f(x) + \sigma \nabla f(x) \nabla f(x)^T \succeq 0. \quad (3.26)$$

It satisfies (3.22) for all $y \neq 0$ if and only if there exists a σ such

$$\nabla^2 f(x) + \sigma \nabla f(x) \nabla f(x)^T \succ 0. \quad (3.27)$$

Hint. We can assume without loss of generality that $\nabla^2 f(x)$ is diagonal.

2. A point $x \in \text{dom } f$ satisfies (3.21) if and only if either $\nabla f(x) = 0$ and $\nabla^2 f(x) \succeq 0$, or $\nabla f(x) \neq 0$ and the matrix

$$H(x) = \begin{bmatrix} \nabla^2 f(x) & \nabla f(x) \\ \nabla f(x)^T & 0 \end{bmatrix}$$

has exactly one negative eigenvalue. It satisfies (3.22) for all $y \neq 0$ if and only if $H(x)$ has exactly one nonpositive eigenvalue.

Hint. You can use the result of part (a). The following result, which follows from the eigenvalue interlacing theorem in linear algebra, may also be useful: If $B \in \mathbb{S}^n$ and $a \in \mathbb{R}^n$, then

$$\lambda_n \left(\begin{bmatrix} B & a \\ a^T & 0 \end{bmatrix} \right) \geq \lambda_n(B).$$

SOLUTION. 记 $B = \nabla^2 f(x)$, $a = \nabla f(x)$ 。二阶拟凸条件 (3.21)、(3.22) 分别为

$$y^T B y \geq 0 \quad (a^T y = 0), \quad y^T B y > 0 \quad (y \neq 0, a^T y = 0).$$

1. 先说明：题中 (3.21) 与 (3.26) 的双向陈述按原样并不正确。例如

$$B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad a = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

满足 $y^T B y = 0$ 对所有 $a^T y = 0$ 成立，但 $B + \sigma a a^T = \begin{bmatrix} \sigma & 1 \\ 1 & 0 \end{bmatrix}$ 对任何 σ 都不是半正定。正确且常用的半正定条件是：若存在 σ 使 $B + \sigma a a^T \succeq 0$ ，则在 $a^T y = 0$ 时有 $y^T B y \geq 0$ 。

严格情形成立。若存在 σ 使 $B + \sigma a a^T \succ 0$ ，则显然 $a^T y = 0, y \neq 0$ 时 $y^T B y > 0$ 。反之，若严格条件成立，且 $a \neq 0$ ，经正交变换可设 $a = (\alpha, 0, \dots, 0)$, $\alpha > 0$ ，并分块

$$B = \begin{bmatrix} \beta & b^T \\ b & C \end{bmatrix}.$$

严格条件等价于 $C \succ 0$ 。此时

$$B + \sigma aa^T = \begin{bmatrix} \beta + \sigma \alpha^2 & b^T \\ b & C \end{bmatrix}$$

在 $\beta + \sigma \alpha^2 - b^T C^{-1} b > 0$ 时正定；取足够大的 σ 即可。若 $a = 0$ ，严格条件就是 $B \succ 0$ ，取任意 σ 即可。

2. 若 $a = 0$ ，(3.21) 即 $B \succeq 0$ ；(3.22) 即 $B \succ 0$ ，矩阵 $H = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ 的非正特征值个数也相应为一个。

下设 $a \neq 0$ 。取正交坐标使 $a = (\alpha, 0, \dots, 0)$ ， $\alpha > 0$ ，并将 B 分块如上。令

$$P = \begin{bmatrix} 1 & 0 & 0 \\ -b/\alpha & I & 0 \\ \beta/\alpha & b^T/\alpha & 1 \end{bmatrix}.$$

直接作合同变换可把

$$H = \begin{bmatrix} \beta & b^T & \alpha \\ b & C & 0 \\ \alpha & 0 & 0 \end{bmatrix}$$

化为

$$P^T H P = \begin{bmatrix} 0 & 0 & \alpha \\ 0 & C & 0 \\ \alpha & 0 & 0 \end{bmatrix}.$$

合同变换保持惯性，而右端由 C 和一个特征值为 $\alpha, -\alpha$ 的 2×2 块组成。因此 H 的负特征值个数等于 C 的负特征值个数加一。(3.21) 等价于 $C \succeq 0$ ，故 H 恰有一个负特征值；反之亦然。(3.22) 等价于 $C \succ 0$ ，故 H 除一个负特征值外其余均为正，即恰有一个非正特征值；反之亦然。

□

PROBLEM X Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, $\text{dom } f$ is convex, and $f(x) > 0$ for all $x \in \text{dom } f$. Show that f is log-concave if and only if for all $x, y \in \text{dom } f$,

$$\frac{f(y)}{f(x)} \leq \exp\left(\frac{\nabla f(x)^T(y-x)}{f(x)}\right).$$

SOLUTION. 令 $g = \log f$ 。由于 $f > 0$ 且可微， g 可微且 $\nabla g(x) = \frac{\nabla f(x)}{f(x)}$ 。 f log-concave 当且仅当 g 凹。对可微函数，凹性等价于一阶上界 $g(y) \leq g(x) + \nabla g(x)^T(y-x)$ ， $\forall x, y$ 。代入 $g = \log f$ 得 $\log f(y) - \log f(x) \leq \frac{\nabla f(x)^T(y-x)}{f(x)}$ 。两边取指数即为题中不等式，反向也由取对数得到。 □

PROBLEM XI [MO79, §3.E.2] **Log-convexity of moment functions.**

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is nonnegative with $\mathbb{R}_+ \subseteq \text{dom } f$. For $x \geq 0$ define

$$\phi(x) = \int_0^\infty u^x f(u) du.$$

Show that ϕ is a log-convex function. (If x is a positive integer, and f is a probability density function, then $\phi(x)$ is the x th moment of the distribution.)

Use this to show that the Gamma function,

$$\Gamma(x) = \int_0^\infty u^{x-1} e^{-u} du,$$

is log-convex for $x \geq 1$.

SKETCH. 对任意 $x, y \geq 0$ 和 $\theta \in [0, 1]$, 由 Holder 不等式,

$$\phi(\theta x + (1 - \theta)y) = \int_0^\infty (u^x f(u))^\theta (u^y f(u))^{1-\theta} du \leq \phi(x)^\theta \phi(y)^{1-\theta}.$$

因此 $\log \phi$ 凸, 即 ϕ log-convex。

取 $f(u) = e^{-u}/u$, 则对 $x \geq 1$,

$$\phi(x) = \int_0^\infty u^x \frac{e^{-u}}{u} du = \int_0^\infty u^{x-1} e^{-u} du = \Gamma(x).$$

由上面的结论, Γ 在 $x \geq 1$ 上 log-convex。 □